



# Pre-Thesis/ Question Research

# Signal Denoising in Finance

Fourier Transforms tell us what freq. is present in a signal, but cannot tell us when they occurred

⇒ Ineffective for analyzing "non-stationary" signals (Market Data)

Solution: (Wavelet Transforms)

Wavelet transf. evaluate a signal using two primary parameters:

- Scale (Dilation): Stretching/Compressing the wavelet.
  - Stretching allows capture of slow, low-freq phenomena
  - Compressing allows to zoom in on rapid, high-freq details
- Position (Translation): Sliding wavelet along entire duration of the signal to identify where specific freq. occur

Continuous Wavelet Transform (CWT):

Def: let  $x(t)$  be a func. at a scale  $a \in \mathbb{R}^{++}$  & translational val.  $b \in \mathbb{R}$ . The CWT of  $x(t)$  is

$$X_w(a, b) = \frac{1}{|a|^{1/2}} \int_{-\infty}^{\infty} x(t) \bar{\psi}\left(\frac{t-b}{a}\right) dt$$

Where  $\psi(t)$  is a cont. func. in both time & freq domain called the mother wavelet. The overline represents operation of complex conj.

To recover original signal  $x(t)$ , the first Inverse Cont. wavelet transform can be exploited:

$$x(t) = C_{\psi}^{-1} \int_0^{\infty} \int_{-\infty}^{\infty} X_w(a, b) \frac{1}{|a|^{1/2}} \bar{\psi}\left(\frac{t-b}{a}\right) db \frac{da}{a^2} \leftarrow \text{first inverse CWT}$$

$\bar{\psi}(t)$  is the dual func. of  $\psi(t)$  &  $C_{\psi} = \int_{-\infty}^{\infty} \frac{\bar{\tilde{\psi}}(\omega) \hat{\tilde{\psi}}(\omega)}{|\omega|} d\omega$ , where hat means Fourier transf. operator.

If  $\bar{\psi}(t) = \psi(t)$  then  $C_{\psi} = \int_{-\infty}^{\infty} \frac{|\hat{\psi}(\omega)|^2}{|\omega|} d\omega \leftarrow \text{wavelet admissible constant}$

Wavelet that satisfies  $0 < C_\psi < \infty$  is called an admissible wavelet  
To recover  $x(t)$  from such a wavelet, use second inverse CWT

$$x(t) = \frac{1}{2\pi \hat{\psi}(1)} \int_0^\infty \int_{-\infty}^\infty \frac{1}{a^2} X_w(a,b) \exp\left(i \frac{t-b}{a}\right) db da$$

Suggesting that  $\psi(t) = w(t) e^{it}$

Discrete Wavelet Transform (DWT):

Def: The DWT of a signal  $x$  is calc by passing it through a low-pass filter  $\leftrightarrow$  impulse  $g$ :

$$y(n) = (x * g)(n) = \sum_{k=-\infty}^{\infty} x(k) g(n-k)$$

The signal is also decomposed simultaneously using a high-pass filter  $h$ .

Since half the freq. of the signal have now been removed, half the sample can be discarded, by Nyquist's sampling thm.

$$\Rightarrow y_{\text{low}}(n) = \sum_{k=-\infty}^{\infty} x(k) g(2n-k)$$

$$y_{\text{high}}(n) = \sum_{k=-\infty}^{\infty} x(k) h(2n-k)$$



## Possible Topics:

### 1. Wavelet denoising & short-horizon ETF predictability

→ Apply wavelet denoising to ETF price series (e.g. SPY, sector ETFs) & test whether signals built on denoised returns have higher predictive power than signals on raw returns.

- Modern control theory?

- CWT, Since stock price series are a cont. time series.

→ Can use CCFT but it's not as reliable (As shown above)

- Stock price series follow causality

2. Comparing denoising methods for ETF signals (MA vs Wavelets vs SSA)  
→ Systematically compare "simple" smoothing (moving avg) vs more advanced denoising (WT, SSA) as preprocessing for ETF trading signals.

• MA: As found in notes & backtested w/ QQQ.

• CWT as above

• Singular Spectral Analysis

→ Singular Value Decomp. (SVD):

$$M = \sum_{i=1}^r \sqrt{\lambda_i} u_i v_i^H, \quad \bullet \text{ } r, \text{ rank of } M$$

•  $u_i, v_i$ ,  $i^{\text{th}}$  eigenvectors of  $MM^H$  &  $M^H M$

•  $\lambda_i$ ,  $i^{\text{th}}$  eigenval of  $MM^H$

Matrix form:  $M = U \Sigma V^H$ ,

•  $\Sigma$ , diagonal matrix of all singular vals in desc. ord.

•  $U$  &  $V$  are matrices containing orthonorm vectors  $u_i$  &  $v_i$

• Components presenting most coherency in the data will be weighted w/ higher singular vals.

⇒ The decomp. of the trajectory matrix in sing. spect. very useful to id trends.

3. Kalman Filter trend extraction for leveraged vs unleveraged ETFs
- Use a Kalman Filter to estimate a "smooth trend" for ETFs & test whether trades against that trend behave differently for leveraged vs unleveraged prod

4. Denoising in ETF pairs trading (spread vs. Kalman-filtered spread)
- Apply denoising to the spread in ETF pairs trading & see whether entry/exit rules based on denoised spreads are superior

5. Lead-lag effects btwn index & sector ETFs

→ Asking "who moves first" - broad market ETFs or sector ETFs & is that lead-lag strong & stable enough to be predictive at short horizon

## 6. Eigen values & Eigenvectors of Stock return Correlation matrix

- To what extent do e.vals & e.vects of the return correlation matrix capture market-wide trend persistence & regime change?
- Can principal components of equity returns be used as early indicators of correlation spikes & market stress?
- Do spectral properties of the cov. matrix improve detection of latent market trends relative to standard return-based indicators?
- Do changes in the leading e.val of the stock return correlation matrix predict market regime shifts or future volatility?

stock ret. Corr. Mat.

S = Stock

|                | S1   | S2   | S3   | S4  | ... | ... | S <sub>n</sub> |
|----------------|------|------|------|-----|-----|-----|----------------|
| S1             | 1.00 |      |      |     |     |     |                |
| S2             |      | 1.00 |      |     |     |     |                |
| S3             |      |      | 1.00 |     |     |     |                |
| S4             |      |      |      | ... |     |     |                |
| ...            |      |      |      |     | ... |     |                |
| ...            |      |      |      |     |     | ... |                |
| ...            |      |      |      |     |     |     | ...            |
| S <sub>n</sub> |      |      |      |     |     |     | 1.00           |

- Compares how correlated returns are from -1 to +1

→ +1 Max Corr.

→ -1 Not Corr.

- e.vals found by  
 $\det(R - \lambda I) = 0$

→ Represent the variance of the data projected onto its principal components

$\lambda_{\max}$ : Strongest underlying trend or cluster in data set

$\lambda_{\min}$ : Indicates redundancy or near-linear dependencies

## ★ 7. Think abt how to connect/link 6. & denoising price-series

→ Does denoising stock return data w/ PCA/eigenvalue filtering improve the stability of trend or volatility estimates?

→ Can removing small eigenvalue components from the return correlation matrix improve out-of-sample trend identification?

**PCA (Principal Component Analysis):** Machine Learning & statistical technique used for dimensionality reduction. It simplifies complex, high-dim datasets while retaining the most important patterns & variance (information)

PCA Condenses the data w/ :

1. **Standardization:** Scales the data so that every variable has a mean of 0 & std deviation of 1.
2. **Covariance Matrix:** It calculates how the variables relate to one another
3. **Eigenvalues & Eigenvectors:** It determines the "Principal Components," which are new, mathematically generated axes
  - PC1 captures largest amount of Variation in data
  - PC2 " " 2<sup>nd</sup> - " " & is perpendicular (uncorrelated) to PC1

**Key Benefits: (PCA)**

- **Data Visualization:** Reduces data to 2D/3D plots  $\Rightarrow$  clusters, trends, & outliers easily visible
- **Noise Reduction:** Discards components that capture least variance
- **Speeds up Algos:** Fewer features  $\Rightarrow$  faster ML training

### Practical Structure:

- Build a correlation or covariance matrix from returns
- Compute eigenvalues & eigenvectors
- Keep only largest components & reconstruct a denoised version of the data
- Compare raw vs denoised series on trend stability, volatility estimates, or forecasting performance

### Further:

→ Does PCA-based denoising of stock returns improve the stability & interpretability of market regime classification?

H1: PCA-denoised data yields more stable regimes over time, w/ fewer erratic regime switches, compared to raw returns

H2: Regime labels from denoised data are more predictive of future volatility or drawdowns than labels from raw data

\* Use Hidden Markov Model for regime analysis. \*

(HMM)



# Steps/Plan

## Turn idea into a question:

Write at least a sentence for each

- Universe: (ex; large-cap U.S. equities / SPY-like index ETFs from 2010-2023)
- Event or Signal: "signal is  $X$  (e.g.  $K$ -day denoised trend, or  $r$ -sigma daily jump)"
- Hypothesis
- Contribution: What does the research tell us / contribute to

## Map out data & methodology:

Data: What Symbols, frequency (daily/intraday), sample period, & where info is gotten from (e.g. yfinance, Polygon, ...)

Preprocessing: Corporate actions, missing data, log returns, volatility estimates, & how outliers will be handled

Experiment design: train/test split, rolling vs fixed window, horizons (1, 3, 5 days)  
eval metrics (Sharpe, hit rate, drawdown,  $t$ -stats)

Trading Layer: entry/exit logic, realistic transaction costs, & simple slippage assumptions so it looks like an actual trading study

Write this as a half-page "Methods plan"

## Literature:

- 3-5 papers that touch core idea
- For each: what data they used, what methodology (filters, models, eval), & what they conclude

Write short "Related Work" section



